

This document provides additional assistance with wiring your Extron IP Link Pro Control Processor to your device. Different components may require a different wiring scheme than those listed below.

For complete operating instructions, refer to the user's manual for the specific IP Link Pro Control Processor or the documentation supplied by the manufacturer of the controlled device.

For more information on using Global Scriptor Modules, refer to the "[Guide to Using Scriptor Modules](#)" document.

Device Specifications

Device Type: Other
Manufacturer: Aprilaire
Firmware Version: N/A
Model(s): 8800

Tested on the Following Software and Firmware Versions

IP Link Pro Control Processor Firmware	Global Scriptor Version
2.06.0002-b002	1.4.1

Version History

Module Version	Date	Notes
1_0_1_0	11/27/2017	Added Permanent Hold functionality.
1_0_0_0	3/7/2017	Initial Version

Module Notes

- Unidirectional variable must be set to 'True' if status is not required. Default value is 'False'.
Example: `InterfaceName.Unidirectional = 'True'`
- connectionCounter variable must be set to the number of queries that will be sent to the device before displaying 'Disconnected' if no response is received. Default value is 15.
Example: `InterfaceName.connectionCounter = 5`
- NodeAddress variable must be set accordingly. Default value is '1'. Node Address ranges from RANGE.
Example: `InterfaceName.NodeAddress = '1'`

Supported Classes and Examples

SerialClass
<code>InterfaceName = ModuleName.SerialClass(ProcessorName, 'COM1', Model='8800')</code>
SerialOverEthernetClass
<code>InterfaceName = ModuleName.SerialOverEthernetClass('192.168.254.254', 2001, Model='8800')</code>

Set Commands

Format with Qualifier:

```
InterfaceName.Set(Command, Value, {'Qualifier Key': 'Qualifier Value'})
```

Format without Qualifier:

```
InterfaceName.Set(Command, Value)
```

Command Alarm	Value 'On'	Value 'Off'	
Qualifier Key 'Node Address'	Qualifier Value '1' – '64'		
Qualifier Key 'Type'	Qualifier Value 'Air Filter Alarm' 'HVAC System Alarm'	Qualifier Value 'Humidifier Alarm'	Qualifier Value 'Dehumidifier Alarm'
# Alarm example InterfaceName.Set('Alarm', 'On', {'Node Address': '1', 'Type': 'Air Filter Alarm'})			
Command ControlMode	Value 'Off' 'Emergency Heat Mode (Heat Pump Only)' 'Dehumidification'	Value 'Heat' 'Auto'	Value 'Cool' 'Humidification'
Qualifier Key 'Node Address'	Qualifier Value '1' – '64'		
# ControlMode example InterfaceName.Set('ControlMode', 'Off', {'Node Address': '1'})			
Command FanMode	Value 'Auto'	Value 'On'	Value 'Circulate'
Qualifier Key 'Node Address'	Qualifier Value '1' – '64'		
# FanMode example InterfaceName.Set('FanMode', 'Auto', {'Node Address': '1'})			
Command HumidifierType	Value 'Flow-Through Type'	Value 'Drain-Less Type'	
Qualifier Key 'Node Address'	Qualifier Value '1' – '64'		
# HumidifierType example InterfaceName.Set('HumidifierType', 'Flow-Through Type', {'Node Address': '1'})			
Command HumidityControlSetpoint	Value 10 to 90 in steps of 1		
Qualifier Key	Qualifier Value		

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'Node Address'	'1' – '64'	
Qualifier Key 'Type'	Qualifier Value 'Humidification'	Qualifier Value 'Dehumidification'
# HumidityControlSetpoint example InterfaceName.Set('HumidityControlSetpoint', 90, {'Node Address': '1', 'Type': 'Humidification'})		
Command PermanentHold	Value 'On'	Value 'Off'
Qualifier Key 'Node Address'	Qualifier Value '1' – '64'	
# PermanentHold example InterfaceName.Set('PermanentHold', 'On', {'Node Address': '1'})		
Command TemperatureScale ¹	Value 'Fahrenheit'	Value 'Celsius'
Qualifier Key 'Node Address'	Qualifier Value '1' – '64'	
# TemperatureScale example InterfaceName.Set('TemperatureScale', 'Fahrenheit', {'Node Address': '1'})		
Command ThermostatSetpoint	Value 4 to 99 in steps of 1	
Qualifier Key 'Node Address'	Qualifier Value '1' – '64'	
Qualifier Key 'Type'	Qualifier Value 'Heat'	Qualifier Value 'Cool'
# ThermostatSetpoint example InterfaceName.Set('ThermostatSetpoint', 99, {'Node Address': '1', 'Type': 'Heat'})		

¹ Range varies on current Temperature Scale. Celsius uses range 4 – 32 and Fahrenheit uses range 40 – 90 when type is 'Heat'. Celsius uses range 6 – 37 and Fahrenheit uses range 42 – 99 when type is 'Cool'.

Status Available

For all commands, call Update to receive the latest status. ConnectionStatus does not support the Update function and is triggered by the device providing a successful response to other Update function calls.

Format with Qualifier:

```
InterfaceName.Update(Command, {'Qualifier Key': 'Qualifier Value'})
Value = InterfaceName.ReadStatus(Command, {'Qualifier Key': 'Qualifier Value'})
InterfaceName.SubscribeStatus(Command, {'Qualifier Key': 'Qualifier Value'}, FeedbackHandler)
FeedbackHandler will be called only when the specified qualifier gets a new status.
```

Format without Qualifier:

```
InterfaceName.Update(Command)
Value = InterfaceName.ReadStatus(Command)
InterfaceName.SubscribeStatus(Command, None, FeedbackHandler)
FeedbackHandler will be called when any qualifier gets a new status.
```

Command Alarm	Value 'On'	Value 'Off'
Qualifier Key 'Node Address'	Qualifier Value '1' – '64'	
Qualifier Key 'Type'	Qualifier Value 'Air Filter Alarm'	Qualifier Value 'Humidifier Alarm'
		Qualifier Value 'Dehumidifier Alarm'
# Alarm examples InterfaceName.Update('Alarm', {'Node Address': '1', 'Type': 'Air Filter Alarm'}) Value = InterfaceName.ReadStatus('Alarm', {'Node Address': '1', 'Type': 'Air Filter Alarm'}) InterfaceName.SubscribeStatus('Alarm', None, FeedbackHandler)		
Command BuiltinHumidity	Value 'String'	
Qualifier Key 'Node Address'	Qualifier Value '1' – '64'	
# BuiltinHumidity examples InterfaceName.Update('BuiltinHumidity', {'Node Address': '1'}) Value = InterfaceName.ReadStatus('BuiltinHumidity', {'Node Address': '1'}) InterfaceName.SubscribeStatus('BuiltinHumidity', None, FeedbackHandler)		
Command ConnectionStatus	Value 'Connected'	Value 'Disconnected'
# ConnectionStatus examples Value = InterfaceName.ReadStatus('ConnectionStatus') InterfaceName.SubscribeStatus('ConnectionStatus', None, FeedbackHandler)		
Command ControlMode	Value 'Off'	Value 'Heat'
	'Emergency Heat Mode (Heat Pump Only)'	'Cool'
		'Humidification'

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	'Dehumidification'		
Qualifier Key 'Node Address'	Qualifier Value '1' – '64'		
# ControlMode examples InterfaceName.Update('ControlMode', {'Node Address': '1'}) Value = InterfaceName.ReadStatus('ControlMode', {'Node Address': '1'}) InterfaceName.SubscribeStatus('ControlMode', None, FeedbackHandler)			
Command ErrorStatus	Value 'No Error' 'Unresponsive Node Error'	Value 'Sensor is Open-Circuited' 'EEPROM Error'	Value 'RH Sensor Error' 'Sensor is Short-Circuited'
Qualifier Key 'Node Address'	Qualifier Value '1' – '64'		
Qualifier Key 'Error Type'	Qualifier Value 'Built-in Temperature Sensor Error' 'Built-in Humidity Sensor Error'	Qualifier Value 'Remote Temperature Sensor 1 Error' 'Local Communication Error'	Qualifier Value 'Remote Temperature Sensor 2 Error' 'EEPROM Error'
# ErrorStatus examples InterfaceName.Update('ErrorStatus', {'Node Address': '1', 'Error Type': 'Built-in Temperature Sensor Error'}) Value = InterfaceName.ReadStatus('ErrorStatus', {'Node Address': '1', 'Error Type': 'Built-in Temperature Sensor Error'}) InterfaceName.SubscribeStatus('ErrorStatus', None, FeedbackHandler)			
Command FanMode	Value 'Auto'	Value 'On'	Value 'Circulate'
Qualifier Key 'Node Address'	Qualifier Value '1' – '64'		
# FanMode examples InterfaceName.Update('FanMode', {'Node Address': '1'}) Value = InterfaceName.ReadStatus('FanMode', {'Node Address': '1'}) InterfaceName.SubscribeStatus('FanMode', None, FeedbackHandler)			
Command HumidifierType	Value 'Flow-Through Type'	Value 'Drain-Less Type'	
Qualifier Key 'Node Address'	Qualifier Value '1' – '64'		
# HumidifierType examples InterfaceName.Update('HumidifierType', {'Node Address': '1'}) Value = InterfaceName.ReadStatus('HumidifierType', {'Node Address': '1'}) InterfaceName.SubscribeStatus('HumidifierType', None, FeedbackHandler)			
Command HumidistatHumidityStatus	Value 'String'		
Qualifier Key 'Node Address'	Qualifier Value '1' – '64'		

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<pre># HumidistatHumidityStatus examples InterfaceName.Update('HumidistatHumidityStatus', {'Node Address': '1'}) Value = InterfaceName.ReadStatus('HumidistatHumidityStatus', {'Node Address': '1'}) InterfaceName.SubscribeStatus('HumidistatHumidityStatus', None, FeedbackHandler)</pre>		
Command HumidityControlSetpoint	Value 10 to 90 in steps of 1	
Qualifier Key 'Node Address'	Qualifier Value '1' – '64'	
Qualifier Key 'Type'	Qualifier Value 'Humidification'	Qualifier Value 'Dehumidification'
<pre># HumidityControlSetpoint examples InterfaceName.Update('HumidityControlSetpoint', {'Node Address': '1', 'Type': 'Humidification'}) Value = InterfaceName.ReadStatus('HumidityControlSetpoint', {'Node Address': '1', 'Type': 'Humidification'}) InterfaceName.SubscribeStatus('HumidityControlSetpoint', None, FeedbackHandler)</pre>		
Command NodeRemoteTemperatureStatus	Value 'String'	
Qualifier Key 'Node Address'	Qualifier Value '1' – '64'	
<pre># NodeRemoteTemperatureStatus examples InterfaceName.Update('NodeRemoteTemperatureStatus', {'Node Address': '1'}) Value = InterfaceName.ReadStatus('NodeRemoteTemperatureStatus', {'Node Address': '1'}) InterfaceName.SubscribeStatus('NodeRemoteTemperatureStatus', None, FeedbackHandler)</pre>		
Command OutdoorTemperatureStatus	Value 'String'	
Qualifier Key 'Node Address'	Qualifier Value '1' – '64'	
<pre># OutdoorTemperatureStatus examples InterfaceName.Update('OutdoorTemperatureStatus', {'Node Address': '1'}) Value = InterfaceName.ReadStatus('OutdoorTemperatureStatus', {'Node Address': '1'}) InterfaceName.SubscribeStatus('OutdoorTemperatureStatus', None, FeedbackHandler)</pre>		
Command PermanentHold	Value 'On'	Value 'Off'
Qualifier Key 'Node Address'	Qualifier Value '1' – '64'	
<pre># PermanentHold examples InterfaceName.Update('PermanentHold', {'Node Address': '1'}) Value = InterfaceName.ReadStatus('PermanentHold', {'Node Address': '1'}) InterfaceName.SubscribeStatus('PermanentHold', None, FeedbackHandler)</pre>		
Command RemoteHumidityStatus	Value 'String'	
Qualifier Key	Qualifier Value	

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'Node Address'	'1' – '64'	
# RemoteHumidityStatus examples InterfaceName.Update('RemoteHumidityStatus', {'Node Address': '1'}) Value = InterfaceName.ReadStatus('RemoteHumidityStatus', {'Node Address': '1'}) InterfaceName.SubscribeStatus('RemoteHumidityStatus', None, FeedbackHandler)		
Command TemperatureScale	Value 'Fahrenheit'	Value 'Celsius'
Qualifier Key 'Node Address'	Qualifier Value '1' – '64'	
# TemperatureScale examples InterfaceName.Update('TemperatureScale', {'Node Address': '1'}) Value = InterfaceName.ReadStatus('TemperatureScale', {'Node Address': '1'}) InterfaceName.SubscribeStatus('TemperatureScale', None, FeedbackHandler)		
Command ThermostatRoomTemperatureControlStatus	Value 'String'	
Qualifier Key 'Node Address'	Qualifier Value '1' – '64'	
# ThermostatRoomTemperatureControlStatus examples InterfaceName.Update('ThermostatRoomTemperatureControlStatus', {'Node Address': '1'}) Value = InterfaceName.ReadStatus('ThermostatRoomTemperatureControlStatus', {'Node Address': '1'}) InterfaceName.SubscribeStatus('ThermostatRoomTemperatureControlStatus', None, FeedbackHandler)		
Command ThermostatSetpoint ¹	Value 4 to 99 in steps of 1	
Qualifier Key 'Node Address'	Qualifier Value '1' – '64'	
Qualifier Key 'Type'	Qualifier Value 'Heat'	Qualifier Value 'Cool'
# ThermostatSetpoint examples InterfaceName.Update('ThermostatSetpoint', {'Node Address': '1', 'Type': 'Heat'}) Value = InterfaceName.ReadStatus('ThermostatSetpoint', {'Node Address': '1', 'Type': 'Heat'}) InterfaceName.SubscribeStatus('ThermostatSetpoint', None, FeedbackHandler)		

¹ Range varies on current Temperature Scale. Celsius uses range 4 – 32 and Fahrenheit uses range 40 – 90 when type is 'Heat'. Celsius uses range 6 – 37 and Fahrenheit uses range 42 – 99 when type is 'Cool'.

Cable and Adapter Requirements

Contact manufacturer for cable and adapter requirements

Notes for the Device

Device requires a RS232 to RS485 converter to operate.

Serial communication

Port Type: RS-232

Baud Rate: 9600

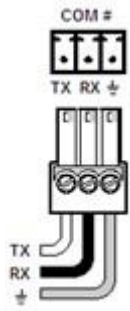
Data Bits: 8

Parity: None

Stop Bits: One

Flow Control: None

Pin Assignments Diagram



Contact Manufacturer for pin assignment diagram and connector type.

Appendix A. Set Commands

Alarm Off Type Air Filter Alarm Node Address 1	SN1 FLTALM=OFF\x0D
Alarm Off Type Air Filter Alarm Node Address 64	SN64 FLTALM=OFF\x0D
Alarm Off Type Dehumidifier Alarm Node Address 1	SN1 DEHALM=OFF\x0D
Alarm Off Type Dehumidifier Alarm Node Address 64	SN64 DEHALM=OFF\x0D
Alarm Off Type Humidifier Alarm Node Address 1	SN1 WPALM=OFF\x0D
Alarm Off Type Humidifier Alarm Node Address 64	SN64 WPALM=OFF\x0D
Alarm Off Type HVAC System Alarm Node Address 1	SN1 SYSALM=OFF\x0D
Alarm Off Type HVAC System Alarm Node Address 64	SN64 SYSALM=OFF\x0D
Alarm On Type Air Filter Alarm Node Address 1	SN1 FLTALM=ON\x0D
Alarm On Type Air Filter Alarm Node Address 64	SN64 FLTALM=ON\x0D
Alarm On Type Dehumidifier Alarm Node Address 1	SN1 DEHALM=ON\x0D
Alarm On Type Dehumidifier Alarm Node Address 64	SN64 DEHALM=ON\x0D
Alarm On Type Humidifier Alarm Node Address 1	SN1 WPALM=ON\x0D
Alarm On Type Humidifier Alarm Node Address 64	SN64 WPALM=ON\x0D
Alarm On Type HVAC System Alarm Node Address 1	SN1 SYSALM=ON\x0D
Alarm On Type HVAC System Alarm Node Address 64	SN64 SYSALM=ON\x0D
Control Mode Auto Node Address 1	SN1 MODE=AUTO\x0D
Control Mode Auto Node Address 64	SN64 MODE=AUTO\x0D
Control Mode Cool Node Address 1	SN1 MODE=COOL\x0D
Control Mode Cool Node Address 64	SN64 MODE=COOL\x0D
Control Mode Dehumidification Node Address 1	SN1 MODE=DEHUM\x0D
Control Mode Dehumidification Node Address 64	SN64 MODE=DEHUM\x0D
Control Mode Emergency Heat Mode (Heat Pump Only) Node Address 1	SN1 MODE=EMHT\x0D
Control Mode Emergency Heat Mode (Heat Pump Only) Node Address 64	SN64 MODE=EMHT\x0D
Control Mode Heat Node Address 1	SN1 MODE=HEAT\x0D
Control Mode Heat Node Address 64	SN64 MODE=HEAT\x0D
Control Mode Humidification Node Address 1	SN1 MODE=HUMID\x0D
Control Mode Humidification Node Address 64	SN64 MODE=HUMID\x0D
Control Mode Off Node Address 1	SN1 MODE=OFF\x0D
Control Mode Off Node Address 64	SN64 MODE=OFF\x0D
Fan Mode Auto Node Address 1	SN1 F=AUTO\x0D
Fan Mode Auto Node Address 64	SN64 F=AUTO\x0D
Fan Mode Circulate Node Address 1	SN1 F=CIRC\x0D
Fan Mode Circulate Node Address 64	SN64 F=CIRC\x0D
Fan Mode On Node Address 1	SN1 F=ON\x0D
Fan Mode On Node Address 64	SN64 F=ON\x0D
Humidifier Type Drain-Less Type Node Address 1	SN1 HUMTYP=1\x0D
Humidifier Type Drain-Less Type Node Address 64	SN64 HUMTYP=1\x0D
Humidifier Type Flow-Through Type Node Address 1	SN1 HUMTYP=0\x0D
Humidifier Type Flow-Through Type Node Address 64	SN64 HUMTYP=0\x0D
Humidity Control Setpoint 10 Type Dehumidification Node Address 1	SN1 SDEH=10%\x0D
Humidity Control Setpoint 10 Type Dehumidification Node Address 64	SN64 SDEH=10%\x0D

Humidity Control Setpoint 10 Type Humidification Node Address 1	SN1 SHUM=10%\x0D
Humidity Control Setpoint 10 Type Humidification Node Address 64	SN64 SHUM=10%\x0D
Humidity Control Setpoint 90 Type Dehumidification Node Address 1	SN1 SDEH=90%\x0D
Humidity Control Setpoint 90 Type Dehumidification Node Address 64	SN64 SDEH=90%\x0D
Humidity Control Setpoint 90 Type Humidification Node Address 1	SN1 SHUM=90%\x0D
Humidity Control Setpoint 90 Type Humidification Node Address 64	SN64 SHUM=90%\x0D
Permanent Hold Off Node Address 1	SN1 PERMHOLD=OFF\x0D
Permanent Hold Off Node Address 64	SN64 PERMHOLD=OFF\x0D
Permanent Hold On Node Address 1	SN1 PERMHOLD=ON\x0D
Permanent Hold On Node Address 64	SN64 PERMHOLD=ON\x0D
Temperature Scale Celsius Node Address 1	SN1 SCALE=C\x0D
Temperature Scale Celsius Node Address 64	SN64 SCALE=C\x0D
Temperature Scale Fahrenheit Node Address 1	SN1 SCALE=F\x0D
Temperature Scale Fahrenheit Node Address 64	SN64 SCALE=F\x0D
Thermostat Setpoint 4 Type Cool Node Address 1	SN1 SC=4\x0D
Thermostat Setpoint 4 Type Cool Node Address 64	SN64 SC=4\x0D
Thermostat Setpoint 4 Type Heat Node Address 1	SN1 SH=4\x0D
Thermostat Setpoint 4 Type Heat Node Address 64	SN64 SH=4\x0D
Thermostat Setpoint 99 Type Cool Node Address 1	SN1 SC=99\x0D
Thermostat Setpoint 99 Type Cool Node Address 64	SN64 SC=99\x0D
Thermostat Setpoint 99 Type Heat Node Address 1	SN1 SH=99\x0D
Thermostat Setpoint 99 Type Heat Node Address 64	SN64 SH=99\x0D

Appendix B. Query Commands

Alarm Type Air Filter Alarm Node Address 1	SN1 FLTALM?\x0D
Alarm Type Air Filter Alarm Node Address 64	SN64 FLTALM?\x0D
Alarm Type Dehumidifier Alarm Node Address 1	SN1 DEHALM?\x0D
Alarm Type Dehumidifier Alarm Node Address 64	SN64 DEHALM?\x0D
Alarm Type Humidifier Alarm Node Address 1	SN1 WPALM?\x0D
Alarm Type Humidifier Alarm Node Address 64	SN64 WPALM?\x0D
Alarm Type HVAC System Alarm Node Address 1	SN1 SYSALM?\x0D
Alarm Type HVAC System Alarm Node Address 64	SN64 SYSALM?\x0D
Built-in Humidity Status Node Address 1	SN1 BIHUM?\x0D
Built-in Humidity Status Node Address 64	SN64 BIHUM?\x0D
Control Mode Node Address 1	SN1 MODE?\x0D
Control Mode Node Address 64	SN64 MODE?\x0D
Error Status Error Type Built-in Humidity Sensor Error Node Address 1	SN1 ERROR?\x0D
Error Status Error Type Built-in Humidity Sensor Error Node Address 64	SN64 ERROR?\x0D
Error Status Error Type Built-in Temperature Sensor Error Node Address 1	SN1 ERROR?\x0D
Error Status Error Type Built-in Temperature Sensor Error Node Address 64	SN64 ERROR?\x0D
Error Status Error Type EEPROM Error Node Address 1	SN1 ERROR?\x0D
Error Status Error Type EEPROM Error Node Address 64	SN64 ERROR?\x0D
Error Status Error Type Local Communication Error Node Address 1	SN1 ERROR?\x0D

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Error Status Error Type Local Communication Error Node Address 64	SN64 ERROR?\x0D
Error Status Error Type Remote Temperature Sensor 1 Error Node Address 1	SN1 ERROR?\x0D
Error Status Error Type Remote Temperature Sensor 1 Error Node Address 64	SN64 ERROR?\x0D
Error Status Error Type Remote Temperature Sensor 2 Error Node Address 1	SN1 ERROR?\x0D
Error Status Error Type Remote Temperature Sensor 2 Error Node Address 64	SN64 ERROR?\x0D
Fan Mode Node Address 1	SN1 F?\x0D
Fan Mode Node Address 64	SN64 F?\x0D
Humidifier Type Node Address 1	SN1 HUMTYP?\x0D
Humidifier Type Node Address 64	SN64 HUMTYP?\x0D
Humidistat Humidity Status Node Address 1	SN1 HUM?\x0D
Humidistat Humidity Status Node Address 64	SN64 HUM?\x0D
Humidity Control Setpoint Type Dehumidification Node Address 1	SN1 SDEH?\x0D
Humidity Control Setpoint Type Dehumidification Node Address 64	SN64 SDEH?\x0D
Humidity Control Setpoint Type Humidification Node Address 1	SN1 SHUM?\x0D
Humidity Control Setpoint Type Humidification Node Address 64	SN64 SHUM?\x0D
Node Remote Temperature Status Node Address 1	SN1 RTS?\x0D
Node Remote Temperature Status Node Address 64	SN64 RTS?\x0D
Outdoor Temperature Status Node Address 1	SN1 OT?\x0D
Outdoor Temperature Status Node Address 64	SN64 OT?\x0D
Permanent Hold Node Address 1	SN1 PERMHOLD?\x0D
Permanent Hold Node Address 64	SN64 PERMHOLD?\x0D
Remote Humidity Status Node Address 1	SN1 OH?\x0D
Remote Humidity Status Node Address 64	SN64 OH?\x0D
Temperature Scale Node Address 1	SN1 SCALE?\x0D
Temperature Scale Node Address 64	SN64 SCALE?\x0D
Thermostat Room Temperature Control Status Node Address 1	SN1 T?\x0D
Thermostat Room Temperature Control Status Node Address 64	SN64 T?\x0D
Thermostat Setpoint Type Cool Node Address 1	SN1 SC?\x0D
Thermostat Setpoint Type Cool Node Address 64	SN64 SC?\x0D
Thermostat Setpoint Type Heat Node Address 1	SN1 SH?\x0D
Thermostat Setpoint Type Heat Node Address 64	SN64 SH?\x0D